



RESEARCH PAPER

Enhanced activities of reproductive system in male rat treated with male silkworm pupae extract

Mu-Gil KWON¹, Deuk-Su KIM², Jeong-Hwan LEE², Se-Ra PARK², Junsik AHN², Joo-Sung KIM³, Kinarm KO^{4,5}, Sungsu PARK^{6,7}, Young-Sun KANG⁸, Young-Kug CHOO¹ and Kisung KO²

1 Department of Biological Science, College of Natural Sciences, Wonkwang University, Iksan, Korea

2 Department of Medicine, Medical Research Institute, College of Medicine, Chung-Ang University, Seoul, Korea

3 Korea Food Research Institute, Gyeong-Gi-Do, Korea

4 Center for Stem Cell Research, SMART Institute of Advanced Biomedical Science, Konkuk University, Seoul, Korea

5 Department of Stem Cell Biology, School of Medicine, Konkuk University, Seoul, Korea

6 Department of Chemistry and Nano Sciences, Ewha Womans University, Seoul, Korea

7 Mechanobiology Institute, National University of Singapore, Singapore

8 Institute of Biomedical Science and Technology, Konkuk University, Seoul, Korea

Correspondence

Kisung Ko, Department of Medicine,
Medical Research Institute, College of
Medicine, Chung-Ang University, 84
Heukseok-Ro, Dongjak-Gu, Seoul 156-756,
Korea.
Email: ksko@cau.ac.kr

Received 27 December 2012;
accepted 31 December 2012.

doi: 10.1111/1748-5967.12011

Abstract

A male silkworm pupae extract with herbal mixtures such as *Rubus coreanus* Miquel, Chinese matrimony vine *Acanthopanax senticosus* and tocopherol can be effectively used to recover or boost stamina. In this study, the effect of the male silkworm pupae extract on the reproductive system of Sprague–Dawley male rats was investigated. No clinical symptoms, and no dying or dead animals were found among experimental male silkworm larvae extract, male silkworm pupae extract and control rat groups during the study. No significant differences were found in body weights, feed or water consumption, or macroscopic examination among the three groups. No lesion was found upon necropsy. However, sperm in the group treated with male silkworm pupae extract were significantly more active than sperm in the control group. Sperm counts in both male groups treated with male silkworm larvae extract and male silkworm pupae extract were significantly higher than sperm counts in the control group. Analysis of phospholipid–hydroperoxide glutathione peroxidase (PHGPx) gene expression showed that the male silkworm pupae extract increased expression in testes by 26.7%. This study shows that the male silkworm pupae extract has potential to be used to enhance the function of the human reproductive system.

Key words: Nuegra, phospholipid–hydroperoxide glutathione peroxidase, silkworm pupae extract, sperm.

Introduction

Nuegra is a natural medicinal product effective for the enhancement of masculine activities without any side effect or toxicity. It is a newly developed product with a substance patented by the rural development administration in South Korea. Nuegra is another product developed based on a home remedy prescribed by “Dong-ui-bo-gam”, a Korean traditional medical book. It is in the form of a soft capsule,

which is easy to take and carry. Therefore, it is a dietary supplement effective for young and middle-aged people suffering from fatigue and weak stamina.

Nuegra is composed of male *Bombyx mori*, *Acanthopanax senticosus*, fruit of omija (*Maximowiczia typical*), *Rubus coreanus* Miquel, L-glutamine, freeze-dried royal jelly, Chinese matrimony vine and tocopherol. According to Dong-ui-bo-gam, a male *B. mori* is effective in the enhancement of masculine activities and is used in dry form after

immediately harvesting male moths as they tear through their cocoons. Dong-ui-bo-gam also says that a male moth starts to mate immediately out of its cocoon and has a lot of stamina to mate with dozens of female moths. Eleutheroside E, a main component of *A. senticosus*, is a lignan that prevents any loss of sexual stamina and helps recovery from any memory loss. According to “Shin-Nong-Bon-Cho-Kyung” written in around 1 AD, Chinese matrimony vine makes muscles and bones stronger with long-period intake in humans (Zhao *et al.* 2010). It also prevents aging, helps to withstand cold and hot weather, supplements stamina and promotes long life.

According to Ryu *et al.* (2002), Nuegra was more effective than Viagra in enhancing stamina. In an animal experiment using mice, the male hormone and nitric oxide (NO) content increased by 32.8% and 16.6%, respectively, in the group treated with Nuegra. This was much higher than 9.9% and 5.4%, respectively, in the group treated with Viagra. The number of sperm increased by 41% in the Nuegra group while it increased by only 11.9% in the Viagra group. In the Nuegra group, athletic endurance increased by 60.0%.

In this study, the effect of male silkworm pupae extract on the reproductive system in Sprague–Dawley (SD) male rats was assessed after a 4-week treatment by oral application. In terms of clinical symptoms, weight change, feed and water consumptions were investigated, and analysis of sperm and sperm counts, measurement of hormones, spermatogenesis-related gene expression analysis and macroscopic or histopathological examination were conducted.

Materials and methods

Composition of male silkworm pupae extract

The product of male silkworm pupae extract used in this study contained soybean oil (42.3%), freeze-dried royal jelly (20.0%), dried extract of male silkworm *B. mori* pupae (18.2%), palm oil (6.0%), dried extract of *A. senticosus* (5.6%), beeswax (2.0%), lecithin (2.0%), natural vitamin E (α -tocopherol) (1.2%), dried extract of *Schizandra chinensis* Baillon (0.9%), dried extract of *R. coreanus* Miquel (0.91%), dried extract of Chinese matrimony vine (0.5%) and *L*-glutamine (0.4%). The sample was stored at 4°C under hermetic seal before use. The product of male silkworm larvae extract was prepared with the same background compounds as above. The control was the background compounds without any extract.

Experiment with SD rats

SD male rats (DBL, Incheon, South Korea) were used. Experimental rats were acclimated for one week before

study, and healthy animals were selected. Based on body weight, rats were randomly selected to have balanced distribution between groups. Male silkworm pupae extract (500 mg/kg body weight) or silkworm extract (250 mg/kg body weight) were orally administered (10 mL/kg body weight) to 7–8 week-old SD rats. The sample was administered once a day using sonde (for oral injection) for the entire experiment (4 weeks). Five rats were housed in each cage (polycarbonate, 26 cm × 42 cm × 18 cm). Room temperature and relative humidity were maintained at $23 \pm 2^\circ\text{C}$ and $50 \pm 10\%$, respectively. The room was ventilated 10–15 times per hour. Light was turned on and off every 12 h, and the light intensity was 150–300 lx. Rat feed (Samyang Oil & Feed Corporation, Youngwale, South Korea) and filter-purified water were provided *ad libitum*.

Observation for symptoms, measurement of body weight and necropsy

After oral administration of silkworm extracts, the subjects were observed for any symptoms. From the next day until day 14, the rats were closely observed once a day for any behavioral changes, intoxication, activities, appearance, autonomic nervous system and death. Body weight was measured just before oral administration, once a week during the study, and one day before necropsy. During necropsy, animals were studied for any significant symptoms. Any organs with suspicious symptoms were fixed in 10% neutral buffered formalin, and a thin-section embedded in paraffin was prepared and stained with hematoxylin–eosin.

Analysis of testes and spermatogenesis

To obtain indirect index for spermatogenesis, the weights of testes on both sides were measured in experimental male rats. The left testis was fixed in Bouin’s solution and embedded in paraffin. After thin-sectioning, the sample was stained with periodic acid Schiff–hematoxylin (PAS–hematoxylin) and histopathological examination was performed. The right testis was added to 12 mL distilled H₂O, homogenized in an Ultra-turrax homogenizer for 2 min at 6000 ×g and then treated with an ultra-sonicator for 3 min at 4–7°C. Homogenization-resistant sperm heads were counted using hemocytometer. The number of sperm heads in 1 g of testis was divided by 6 days to yield the daily sperm production (DSP) rate per gram of testis. To study the number of sperm in tails of the epididymis, a left tail of the epididymis was removed and weighed. Then, the tissue was homogenized and the number of homogenization-resistant sperm heads remaining in the tail of the epididymis was counted using a hemocytometer.

Analysis of sperm in seminal duct for morphology and motility

A left seminal duct was excised and added to 3 mL of modified Tyrode's media in Petri dishes, and incubated at 37°C, 5% CO₂ for 10 min. After the tissue was removed from the media, 0.1 mL of the remaining medium was mixed in a 1:1 volume ratio (2% eosin: 3% citric acid) to prepare a smear sample. One hundred sperm per sample were observed under a microscope for any morphological significance. Some media samples were analyzed using a semen analyzer (SAIS Medical Supply, Gangwon-Do, Korea) for motility (%), hyperactivity (%), curvilinear velocity (VCL) (µm/sec), straight line velocity (µm/sec), average path velocity (µm/sec), linearity (%), straightness, beat cross frequency (Hz), mean angular displacement dance, dance-mean and lateral head displacement (µm). This is the measure of a sperm head along the straight line between its starting point to its last detection point. VCL is the average velocity measured over the point-to-point track followed by the cell. Time-average velocity (VAP) measures the sperm head along its spatial average trajectory (Penfold *et al.* 1998; Schmidt & Kamp 2004; Slotter *et al.* 2006).

Measurement of blood testosterone level

Male blood testosterone levels are indicators for spermatogenesis and the function of testis. Therefore, blood samples were collected from veins of male SD rats. Blood serum was obtained by centrifugation of the blood sample at 2000 ×g for 10 min. The serum sample was analyzed with Y-count (COBRA 5010 Quantum, Meriden, CT, USA) using a free testosterone radioimmunoassay (RIA) kit (Dslactive free testosterone coated RIA kit, Diagnostic system laboratories Inc., Webster, TX, USA) with the competitive RIA and coated tube method.

Analysis of gene expression of PHGPx in testes of rats

Phospholipid-hydroperoxide glutathione peroxidase (PHGPx) gene encoding for a Se-dependent antioxidant

enzyme is highly expressed in male germ cells and plays an important role in production of sperm. Therefore, total RNA was purified from the testes of rats using TRIzol, and the amount of RNA was measured with spectrophotometry at 260 nm. First-strand cDNA was synthesized from total RNA, 2 µg using Superscript Preamplification System Synthesis (Life Technologies Ltd, Paisley, UK). The target cDNA fragment was amplified using thermal cycler (MJ Research, Inc., San Francisco, CA, USA) as follows: 1 cycle of 95°C for 3 min, 35 cycles of 95°C for 1 min (denaturing), 55°C for 1 min (annealing), 72°C for 1.5 min (extension), and finally 1 cycle of 72°C (extension) for 5 min. Polymerase chain reaction (PCR) product was loaded onto 1% agarose gel and the PHGPx gene expression was confirmed and the expression density in testes was analyzed.

Statistical analysis

One-way analysis of variance (ANOVA) was used to study if there was any statistically significant difference in weights between groups. When the *F*-value was significant, Dunnett's *T*-test was used to compare each group to control group. χ^2 analysis was used for frequencies of lesion occurrence.

Results and discussion

During the study, no clinical symptoms were observed across all groups. No dead or dying animals were found during the period. In addition, during the experimental period, no significantly different body weight was found in all experimental groups compared to the control group. On day 0, the mean body weight was around 240 g. On day 28, the mean body weight increased to around 360 g among three groups. Feed and water consumption was generally irregular during the experimental period in all groups studied (Table 1). Although the feed and water consumption increased over time in all groups, no significant difference was found among groups (Table 1). Male pupae extract enhanced the testosterone level in the serum (Ryu *et al.* 2002). However, in this study the testosterone level was significantly different among groups (data not shown),

Table 1 Food and water consumption in SD rats administered silkworm and male silkworm pupae extracts orally for 4 weeks

Treatment	Week 1		Week 2		Week 3		Week 4	
	Food (g)	Water (g)	Food (g)	Water (g)	Food (g)	Water (g)	Food (g)	Water (g)
Control (0 mg/kg)	146.0	295.4	176.6	316.3	198.0	289.2	205.1	500.3
Male silkworm larvae extract (250 mg/kg)	175.1	330.1	190.7	327.2	190.6	347.4	217.6	426.4
Male silkworm pupae extract (500 mg/kg)	160.8	235.3	178.7	339.8	204.6	328.3	208.8	489.5

Results are mean value (n = 8).

which is consistent with a previous study of Oh *et al.* (2012). In addition, no significant difference was found among groups in all absolute or relative organ weights (Table 2).

The sperm counts in the testes and the epididymis total sperm counts in both groups treated with male silkworm larvae extract and male silkworm pupae extract were significantly higher than that of the control group (Table 3).

The sperm activity of the group treated with male silkworm larvae extract was significantly higher than the control group in percent motility (MOT), which can be calculated by the number of sperm with time-average velocity divided by the total number of sperm analyzed times 100, as well as curvilinear line velocity (VCL) and straight line velocity (VSL) (Wang *et al.* 1991; Álvarez *et al.* 2003;

Table 2 Absolute and relative organ weights of male SD rats administered with silkworm pupae extracts for 4 weeks

Item	Group					
	Absolute organ weights (g)			Relative organ weights (g)		
	Control	Male silkworm larvae extract	Male silkworm pupae extract	Control	Male silkworm larvae extract	Male silkworm pupae extract
Final weights	357.10 ±14.982	357.81 ±12.364	354.23 ±9.094	357.10 ±14.982	357.81 ±12.364	354.23 ±9.094
Testis-L	1.79 ±0.08	1.74 ±0.09*	1.80 ±0.08	0.050 ±0.002	0.049 ±0.003	0.050 ±0.002
Testis-R	1.81 ±0.087	1.78 ±0.121	1.81 ±0.018	0.050 ±0.002	0.050 ±0.004	0.050 ±0.002
Epididymis-L	0.56 ±0.048	0.52 ±0.034	0.55 ±0.088	0.016 ±0.002	0.015 ±0.001	0.016 ±0.002
Epididymis-R	0.56 ±0.025	0.54 ±0.040	0.55 ±0.052	0.016 ±0.001	0.015 ±0.001	0.016 ±0.001
Prostate gland	0.90 ±0.493	0.62 ±0.229	0.65 ±0.277	0.025 ±0.013	0.018 ±0.006	0.018 ±0.008
Pituitary gland	0.014 ±0.003	0.012 ±0.004	0.013 ±0.005	0.0004 ±0.0001	0.0003 ±0.00001	0.0004 ±0.0001
Brain	1.92 ±0.081	1.90 ±0.133	1.91 ±0.065	0.054 ±0.002	0.054 ±0.003	0.053 ±0.002
Thymus	0.52 ±0.093	0.53 ±0.082	0.56 ±0.101	0.015 ±0.002	0.015 ±0.002	0.016 ±0.003
Heart	1.23 ±0.068	1.24 ±0.076	1.27 ±0.113	0.035 ±0.003	0.035 ±0.002	0.035 ±0.003
Lung	1.59 ±0.121	1.51 ±0.076	1.61 ±0.170	0.045 ±0.003	0.043 ±0.003	0.045 ±0.004
Liver	12.91 ±0.921	12.30 ±1.152	12.86 ±0.521	0.362 ±0.020	0.347 ±0.030	0.360 ±0.010
Spleen	0.81 ±0.086	0.80 ±0.078	0.78 ±0.053	0.023 ±0.002	0.023 ±0.002	0.022 ±0.001
Kidney-L	1.16 ±0.10	1.11 ±0.083	1.16 ±0.074	0.033 ±0.001	0.031 ±0.002	0.033 ±0.002
Kidney-R	0.97 ±0.11	1.15 ±0.057	1.15 ±0.077	0.033 ±0.001	0.032 ±0.002	0.032 ±0.001
Adrenal gland-L	0.033 ±0.007	0.032 ±0.005	0.034 ±0.005	0.0009 ±0.0002	0.0009 ±0.0001	0.0010 ±0.0001
Adrenal gland-R	0.030 ±0.004	0.034 ±0.009	0.031 ±0.003	0.0008 ±0.0001	0.0010 ±0.0002	0.0009 ±0.0001
Thyroid gland-L	0.010 ±0.005	0.011 ±0.005	0.013 ±0.003	0.0003 ±0.0002	0.0003 ±0.0001	0.0004 ±0.001
Thyroid gland-R	0.010 ±0.003	0.011 ±0.005	0.012 ±0.004	0.0003 ±0.0001	0.0003 ±0.0001	0.0003 ±0.001
Submaxillary gland	0.75 ±0.099	0.71 ±0.105	0.72 ±0.082	0.023 ±0.003	0.020 ±0.003	0.020 ±0.002

Table 3 Sperm counts in SD rats administered orally with male silkworm larvae extract and male silkworm pupae extract (Nuegra) for 4 weeks

Treatment	Items		
	Count	Count/g	Sperm product/day
Control (0 mg/kg)	8.86×10^7	4.91×10^7	8.05×10^6
Male silkworm larvae extract (250 mg/kg)	9.38×10^7	5.23×10^7	8.58×10^6
Male silkworm pupae extract (500 mg/kg)	9.62×10^7	$5.44 \times 10^{7*}$	8.92×10^6

Results are mean value ($n = 8$).
Significant differences as compared with control $*P < 0.05$.

Figure 1 Sperm analysis in SD rats administered orally with Nuegra for 4 weeks. MOT, motility (%); HYP, hypermotility (%); VCL, curvilinear line velocity ($\mu\text{m/s}$); VSL, straight line velocity ($\mu\text{m/s}$); VAP, average path velocity ($\mu\text{m/s}$); LIN, linearity (%); STR, straightness (%); BCF, beat cellular frequency (Hz); MAD, mean angular displacement ($^\circ$); WOB, wobble (%); DNM, dance mean (μm); ALH, lateral head displacement (μm). Results are mean $\pm$ SD ($n = 8$). Significant differences as compared with control: $*P < 0.05$.

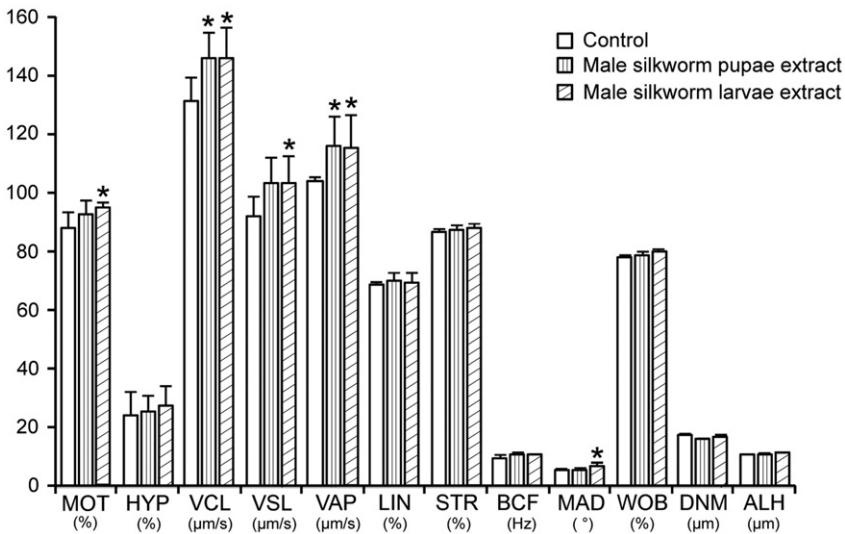
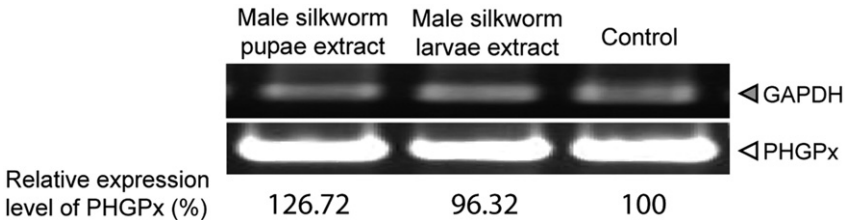


Figure 2 Expression of specific transcripts of *PHGPx* in testis of rats treated with Nuegra for 4 weeks. Relative expression level of *PHGPx* (%) was calculated by the formula $PHGPx/GAPDH \times 100$.



Abascal *et al.* 2007; Tubbs & Thomas 2009). The group treated with male silkworm pupae extract was significantly more active than the control group in VCL and VAP (Fig. 1).

The expression level of *PHGPx* gene in testicular tissues treated with male silkworm pupae extract was about 26.7% higher than that in the control group (Fig. 2). *PHGPx* is a distinct gene product of at least 155 amino acids. Mature *PHGPx* is present mostly in the mitochondria, nucleus and cytoplasm. The enzyme is highly expressed in rat testis, which indicates that it is responsible for mitochondria of cells undergoing differentiation to spermatozoa. The activity of *PHGPx* was detectable after puberty of the other animal

species such as rat (Sunde *et al.* 1993; Roveri *et al.* 1994; Dash *et al.* 2006).

In autopsy findings at the end of study, no visible lesion by sample treatment was observed for all experimental animals (data not shown).

In the histopathological study, specimens from all organ tissues were prepared and observed under an optical microscope. No lesion was observed by the sample treatment. However, the number of sperm in the testis or epididymis was higher in the group treated with male silkworm pupae extract than the control group (data not shown).

In modern societies, interest in health and sexual activities has been constantly increasing. Due to their minimal side

effects, herbs as medicines and natural food ingredients have been popular and are used to develop stamina-boosting products. A male silkworm pupae extract was developed as a herbal mixture product made of male silkworms and natural ingredients. It is a dietary supplement effective in reducing diabetes and boosting stamina.

Many natural ingredients added to the male silkworm pupae extract provide various health-promoting benefits (Ratcliffe *et al.* 2011). A male silkworm pupae as a component of the male silkworm pupae extract stimulates human male libido and prevents involuntary emission of semen, nocturnal emissions or bloody urine. It keeps the feet warm, increases energy and strengthens the erection. Freeze-dried royal jelly is effective for fatigue and enhances stamina. Huang *et al.* 2011 reports *A. senticosus* is an energy-boosting natural substance effective for human male impotence. In addition, *Schizandra chinensis* Baillon is effective in healing damaged intestinal organs. Chinese matrimony vine is effective for nourishment of intestinal organs and healing diabetes (Harunobu & Farnsworth 2011). *Rubus coreanus* Miquel enhances sexual performance (Bae *et al.* 2007; Yoon *et al.* 2007). Ginseng is commonly used to heal nervous prostration and impotence (Chan 2012). *Cornus officinalis* enhances kidney function and reduces blood pressure (Huahong & Zhezhi 2008). All of these ingredients including vitamin E and L-glutamine are natural with few side effects.

In this study, no significant difference was found among groups in terms of clinical symptoms, weight change, feed and water consumption and macroscopic examination. However, the group treated with male silkworm pupae extract had significantly higher sperm activities than the control group.

In addition, the number of sperm in the group treated with male silkworm pupae extract was significantly higher than the number of sperm in the control group. The gene expression level of *PHGPx* in the testes of group treated with male silkworm pupae extract was higher by 26.7% than the gene expression level in the testes of control group.

Taken together, this study indicates that male silkworm pupae extract enhanced activities of male rat reproductive systems, particularly sperm production and several sperm activities, without any clinical side effects in male rats with male silkworm larvae extract. Our current studies are more advanced than the previous research (Ryu *et al.* 2002; Oh *et al.* 2012) where only the effect of male silkworm pupa powder on erectile dysfunction and testosterone levels in rats were studied without comparing the male silkworm larvae extract.

Acknowledgements

This research was supported by a grant from Korea Institute of Planning and Evaluation for Technology of Food, Agricul-

ture, Forestry and Fisheries (iPET-Code# 111096-03-1-SB010); National Research Foundation of Korea Grant funded by the Korean Government (MEST) (NRF-2012-0026969); the Next-Generation BioGreen 21 Program (SSAC, grant# PJ00906), Rural Development Administration; Research Institute for Biomedical and Pharmaceutical Sciences funded by the Health Care System, Chung-Ang University in 2012.

References

- Abascal F, Cosson J, Christian F (2007) Characterization of sperm motility in sea bass: the effect of heavy metals and physicochemical variables on sperm motility. *Journal of Fish Biology* **70**: 509–522.
- Álvarez C, Castilla JA, Martínez L, Ramírez JP, Vergara F, Gaforio JJ (2003) Biological variation of seminal parameters in healthy subjects. *Human Reproduction* **18**: 2082–2088.
- Bae JY, Lim SS, Choi JS, Kang YH (2007) Protective actions of *Rubus coreanus* ethanol extract on collagenous extracellular matrix in ultraviolet-B irradiation-induced human dermal fibroblasts. *Nutrition Research and Practice* **1**: 279–284.
- Chan SW (2012) *Panax ginseng*, *Rhodiola rosea* and *Schisandra chinensis*. *International Journal of Food* **1**: 75–81.
- Dash B, Metz R, Huebner HJ, Porter W, Phillips TD (2006) Molecular characterization of phospholipid hydroperoxide glutathione peroxidases from *Hydra vulgaris*. *Gene* **381**: 1–12.
- Harunobu A, Farnsworth NR (2011) A review of botanical characteristics, phytochemistry, clinical relevance in efficacy and safety of *Lycium barbarum* fruit (Goji). *Food Research International* **44**: 1702–1717.
- Huahong W, Zhezhi W (2008) Comparative determination of ursolic acid and oleanolic acid of *Macrocarpum officinalis* (Sieb. et Zucc.) Nakai by RP-HPLC. *Industrial Crops and Products* **28**: 328–332.
- Huang LZ, Huang BK, Ye Q, Qin LP (2011) Bioactivity-guided fractionation for anti-fatigue property of *Acanthopanax senticosus*. *Journal of Ethnopharmacology* **133**: 213–219.
- Oh HG, Lee HY, Kim JH, Kang YR, Moon DI, Seo MY *et al.* (2012) Effects of male silkworm pupa powder on the erectile dysfunction by chronic ethanol consumption in rats. *Laboratory Animal Research* **28**: 83–90.
- Penfold LM, Holt C, Holt WV, Welch GR, Cran DG, Johnson LA (1998) Comparative motility of X and Y chromosome-bearing bovine sperm separated on the basis of DNA content by flow sorting. *Molecular Reproduction and Development* **50**: 323–327.
- Ratcliffe NA, Mello CB, Garcia ES, Butt TM, Azambuja P (2011) Insect natural products and processes: new treatments for human disease. *Insect Biochemistry and Molecular Biology* **41**: 747–769.
- Roveri A, Maiorino M, Nisii C, Ursini F (1994) Purification and characterization of phospholipid hydroperoxide glutathione peroxidase from rat testis mitochondrial membranes. *Biochimica et Biophysica Acta* **1208**: 211–221.

- Ryu KS, Ahn MY, Lee HS, Kim IS, Kim JW, Kim SH *et al.* (2002) The tonic of the extract from male silkworm (*Bombyx mori* L.) pupae on rats. *International Journal of Industrial Entomology* **5**: 123–126.
- Schmidt H, Kamp G (2004) Induced hyperactivity in boar spermatozoa and its evaluation by computer-assisted sperm analysis. *Reproduction* **128**: 171–179.
- Sloter E, Schmid TE, Marchetti F, Eskenazi B, Nath J, Wyrobek AJ (2006) Quantitative effects of male age on sperm motion. *Human Reproduction* **21**: 2868–2875.
- Sunde RA, Dyer JA, Moran TV, Evenson JK, Sugimoto M (1993) Phospholipid hydroperoxide glutathione peroxidase: full-length pig blastocyst cDNA sequence and regulation by selenium status. *Biochemical and Biophysical Research Communications* **193**: 905–911.
- Tubbs C, Thomas P (2009) Progesterone signaling through an olfactory G protein and membrane progesterone receptor-alpha in Atlantic croaker sperm: potential role in induction of sperm hypermotility. *Endocrinology* **150**: 473–484.
- Wang C, Leung A, Tsoi WL, Leung J, Ng V, Lee KF *et al.* (1991) Evaluation of human sperm hyperactivated motility and its relationship with the zona-free hamster oocyte sperm penetration assay. *Journal of Andrology* **12**: 253–257.
- Yoon HJ, Kim CS, Lee KY, Yang SY (2007) Antioxidant activity of *Rubus coreanus* fruit extract: in comparison to green tea extract. *Chonnam Medical Journal* **46**: 148–155.
- Zhao WG, Chung JW, Cho YI, Rha WH, Lee GA, Ma KH *et al.* (2010) Molecular genetic diversity and population structure in *Lycium* accessions using SSR markers. *Molecular biology and Genetics* **333**: 793–800.